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Suicidal Outcomes Among Adolescent Females: Exploration of Important Risk and Protective Factors

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Background

- Suicide is an escalating public health issue among youth aged 10 to 24 years old in the United States.
- Female adolescents are experiencing rapidly increasing rates of suicide compared to males.

Methods

Dataset: 2023 National Youth Risk Behavior Survey; female adolescents only (n = 9,884).

Analysis: ML workflow to classify suicidal outcomes combined with a custom model comparison procedure to optimize performance while reducing disparities using a grid search on processing methods.

Step 1: Train and Evaluate a Set of Models

Processed

Pre-Processing	Post-Processing
Dropping Subgroup Label	Threshold Optimization
Outcome Balancing Weights	Threshold Optimization
None	Threshold Optimization
Dropping Subgroup Label	None
Outcome Balancing Weights	None

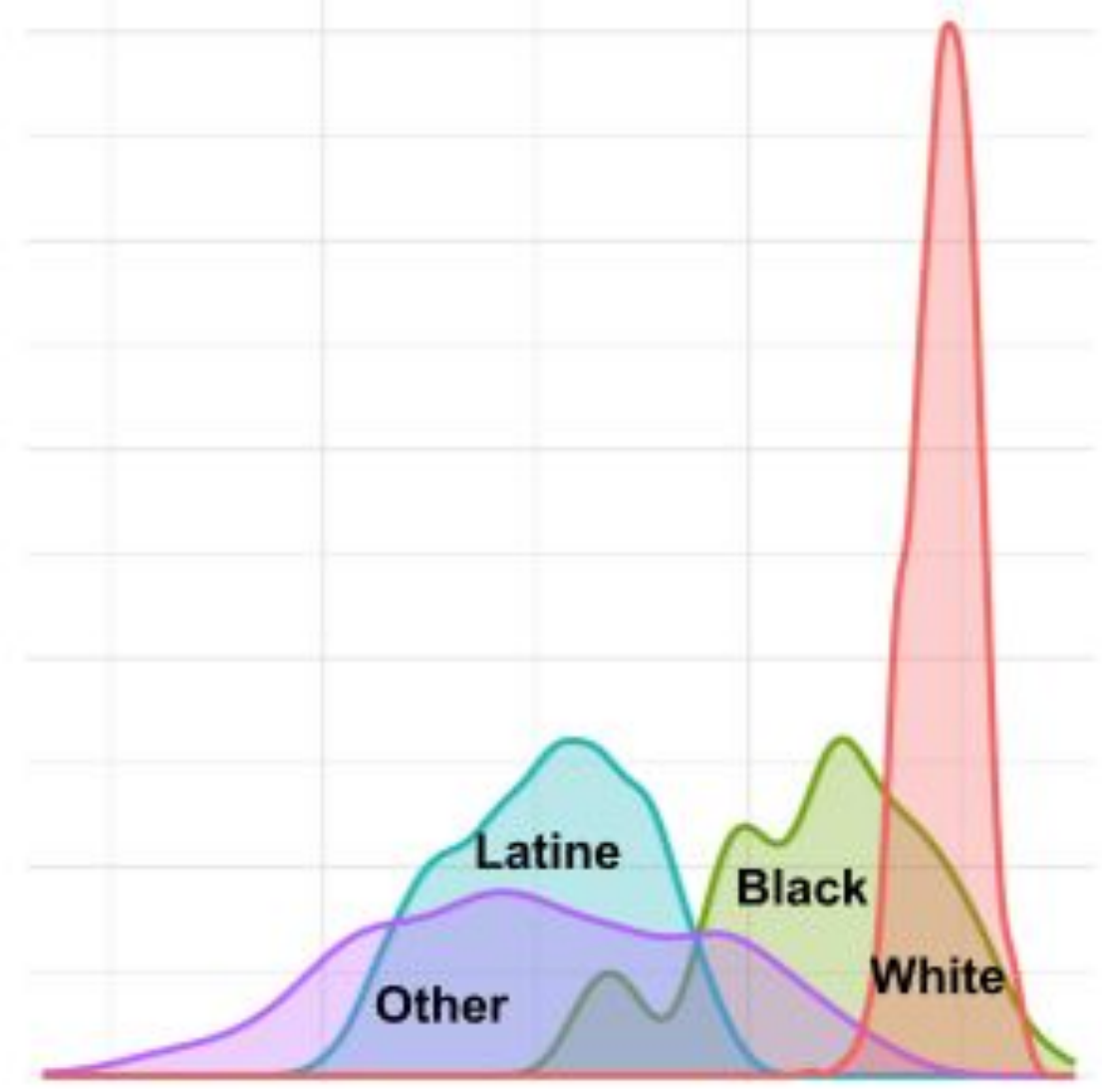
No Processing

Pre-Processing	Post-Processing
None	None

Step 2: Optimize for Performance (F1 Score)

1) Select Processed Models:
Processed - No Processing = F1 Score > 0 or F1 Score = 0
2) Select No Processing Models

Step 3: Model Comparison to Optimize for Least Subgroup Disparities (Disparities Score)



$$\text{Dis Score} = \sum_{i=1}^{\text{Model Metrics (M)}} \frac{\text{Number of group differences}_i}{\text{Number of group pairwise comparisons}_i}$$

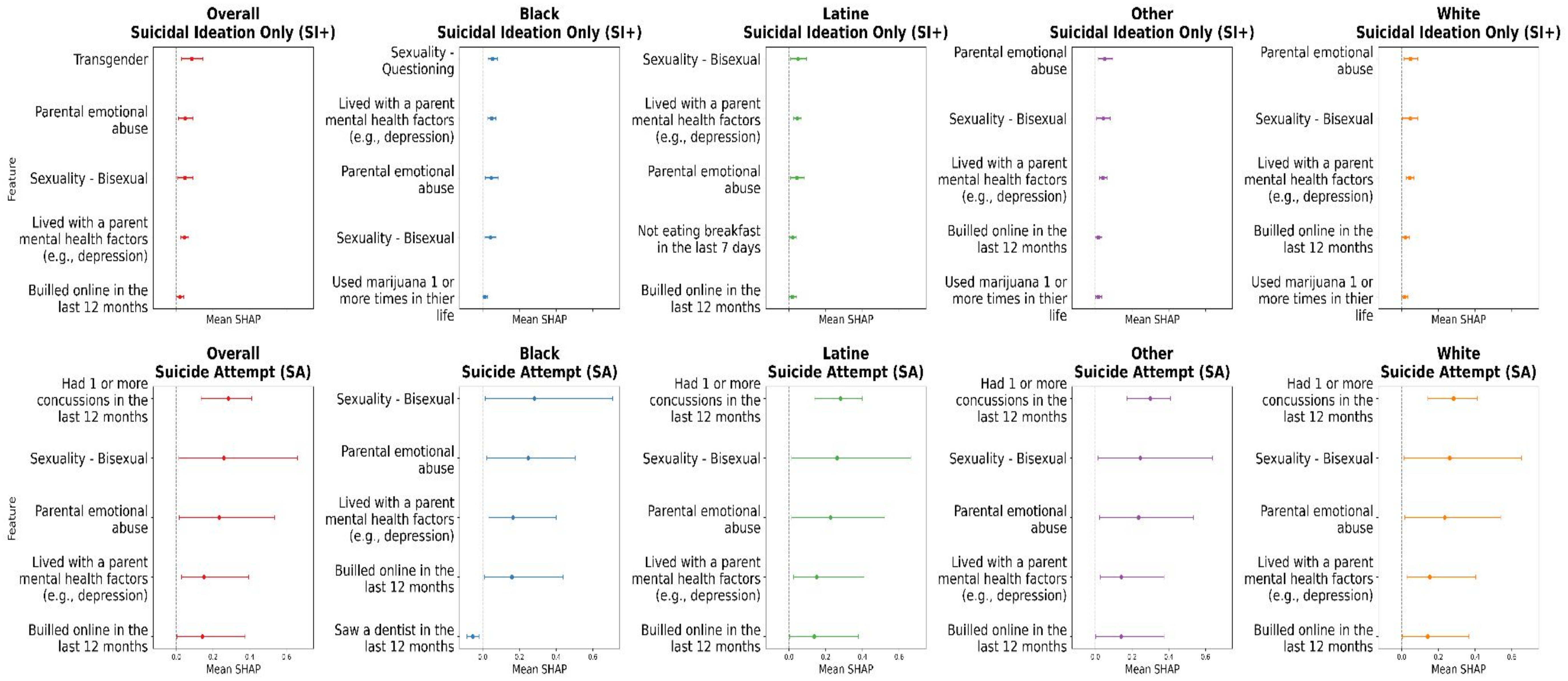
Results

The best models were found for suicidal ideation (Random Forest) and suicide attempt (XGBoost with outcome balancing weights) which displayed subgroup disparities.

Outcome	F1 Score	Accuracy	Brier Score	AUC-PRC	Dis. Score*
Ideation	0.32 (0.30,0.34)	0.62 (0.60,0.64)	0.11 (0.10,0.12)	0.25 (0.23,0.27)	0.93
Attempt	0.34 (0.30,0.40)	0.72 (0.69,0.74)	0.12 (0.11,0.13)	0.39 (0.34,0.45)	0.93

*Dis. Score: Disparity Score which includes subgroup comparisons of AUC-ROC, AUC-PRC, F1 Score, Accuracy, Precision, Recall, and Brier Score

SHAP Feature Importance Overall and By Race/Ethnicity by Outcome



Mean SHAP values with 95% confidence intervals (significant findings only)

Key Takeaways

- Key risk and protective factors can directly inform tailored prevention efforts for adolescent females.
- Sexuality, and parental factors such as mental health, emotional abuse, and substance abuse were important across groups and outcomes
- Concussions and vape usage were risk factors unique to suicide attempts
- One potential protective factor emerged for Black female adolescents (“saw a dentist”) which might be a proxy for parental support

Limitations

- There were a limited number of potential protective factors measured in the Youth Risk Behavior Survey
- Further reducing subgroup disparities in model performance should be explored (pre, in, and post processing methods)

Github

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